



Mini Airline

Reservation System

Software Requirements & System Design Document

Version 1.0 • 2025

Built with ASP.NET Core | Entity Framework | SQL Server

An internal flight management platform enabling passengers to search, book, and manage tickets while earning reward points — powered by automated distance-based pricing.

01 System Overview

The **Mini Airline Reservation System** is a domestic flight management platform that allows passengers to search for flights, book tickets, calculate ticket prices based on distance, and earn reward points. Administrators can manage flights, users, and bookings through a dedicated dashboard.

Goal	Description
Simplify Booking	Enable passengers to search and book domestic flights easily.
Flight Management	Admins can add, update, and remove flights in real time.
Automated Pricing	Ticket price is calculated automatically based on distance.
Rewards Program	Passengers earn loyalty points redeemable for discounts.

02 Stakeholders

Stakeholder	Role
Passenger	Primary user who searches and books flights.
Admin	Manages flights, users, bookings, and the rewards system.
Airline Staff	Optional actor for flight status updates and check-in assistance.

03 System Actors

■ Passenger

- Register an account
- Log in / Log out
- Search for available flights
- Book a ticket
- Cancel a reservation
- View reward points balance
- View personal booking history

■ Admin

- Add new flights
- Update existing flights
- Delete flights

- Manage user accounts
- View all bookings
- Manage the rewards points system

04 Functional Requirements

■ 4.1 User Management

- User Registration — new users can create an account.
- User Login / Logout — secure authentication.
- View Profile — passengers can view their information.
- Update Profile — passengers can update their personal details.

■ 4.2 Flight Management (Admin)

Admins can perform full CRUD operations on flights. Each flight record contains:

Field	Type	Description
Flight Number	String	Unique identifier for the flight
Departure City	String	Origin city
Destination City	String	Destination city
Distance	Float	Distance in kilometers
Departure Time	DateTime	Scheduled departure date and time
Arrival Time	DateTime	Scheduled arrival date and time
Total Seats	Integer	Total seat capacity
Available Seats	Integer	Remaining seats available for booking
Base Price	Decimal	Base ticket price (before tax/class modifier)

■ 4.3 Flight Search

Users can search flights using:

- Departure City
- Destination City
- Travel Date

Search results display:

- Available flights matching the criteria
- Calculated ticket price
- Number of remaining seats

■ 4.4 Ticket Booking

Passengers can:

- Select a flight from search results

- Specify the number of seats required
- Confirm and finalize the booking

Upon confirmation the system will:

- Decrease the available seat count
- Generate a Ticket record
- Calculate and record the final price

■ 4.5 Ticket Price Calculation

Price Formula:

Ticket Price = Distance × Price per KM

Example:

Distance = 500 km

Price per KM = \$2

Ticket Price = \$1,000

Optional modifiers: taxes, seat class (Economy / Business / First).

■ 4.6 Rewards System

Points Formula:

Points Earned = Distance ÷ 10

Example:

Distance = 500 km → 50 Points

Accumulated points can be redeemed as discounts on future ticket purchases.

■ 4.7 Booking History

Passengers can view their complete booking history, including:

- Flight details (route, date, time)
- Ticket price paid
- Booking status (Confirmed / Cancelled)

05 Non-Functional Requirements

Category	Requirement
Performance	Flight search results must be returned within acceptable response times.
Security	All endpoints protected via Authentication & Role-based Authorization.

Scalability	System architecture must support growth in users and flights.
Reliability	Double-booking prevention — the system must never allow overselling seats.

06 System Features

✈️ Core Features	■ Advanced Features
• Flight Search • Ticket Booking • Distance-Based Pricing • Rewards System • Booking Management	• Seat Selection • Flight Status Tracking • Email Confirmation • Cancel Ticket • Discount Coupons • Loyalty Levels (Silver / Gold / Platinum)

07 Database Design

Users	Definition
Id	INT PRIMARY KEY
Name	NVARCHAR(100)
Email	NVARCHAR(150) UNIQUE
Password	NVARCHAR(255) — hashed
RewardPoints	INT DEFAULT 0

Flights	Definition
Id	INT PRIMARY KEY
FlightNumber	NVARCHAR(20) UNIQUE
DepartureCity	NVARCHAR(100)
DestinationCity	NVARCHAR(100)
Distance	FLOAT
DepartureTime	DATETIME
ArrivalTime	DATETIME
TotalSeats	INT
AvailableSeats	INT
BasePrice	DECIMAL(10,2)

Tickets	Definition
Id	INT PRIMARY KEY
UserId	INT FK -> Users

FlightId	INT FK -> Flights
SeatNumber	NVARCHAR(10)
Price	DECIMAL(10,2)
BookingDate	DATETIME
Status	NVARCHAR(20) — Confirmed/Cancelled

Rewards	Definition
Id	INT PRIMARY KEY
UserId	INT FK -> Users
Points	INT
TransactionDate	DATETIME
Description	NVARCHAR(200)

08 Technology Stack

Layer	Technology	Purpose
Backend	ASP.NET Core	REST API / MVC web application
ORM	Entity Framework Core	Database access and migrations
Database	SQL Server	Persistent data storage
Frontend	Razor Pages / MVC	Server-side rendered UI
CSS Framework	Bootstrap 5	Responsive design
Authentication	ASP.NET Identity	User registration, login, roles

09 API Endpoints

Method	Endpoint	Description
GET	/flights	Retrieve all available flights
GET	/flights/search	Search flights by city and date
POST	/booking	Create a new ticket booking
GET	/user/bookings	Get current user's booking history
POST	/cancel	Cancel an existing booking
GET	/rewards	Get current user's reward points

10 Possible Future Improvements

Feature	Details
Payment Integration	Integrate payment gateways (Stripe, PayPal) for online transactions.
Real Airline API	Connect to live airline data feeds for real-time schedules.
Mobile Application	Develop a cross-platform mobile app (iOS & Android).
AI Price Prediction	Use machine learning to dynamically adjust and predict ticket prices.
